

Crater Counting on Charon

I counted the impact craters on Pluto's moon, Charon. This was done to create a heat map of crater density which in turn would show which areas of the surface have been relatively recently resurfaced. This would allow us to determine where Charon potentially has higher amounts of tectonic activity.

To gather my data I used a map from <https://astrocloud.wr.usgs.gov/index.php?view=map2> using https://astrogeology.usgs.gov/search/map/Charon/NewHorizons/Charon_NewHorizons_GlobalMosaic_300m_Jul2017. I then applied a grid over the map and blacked out the areas that were too low resolution to easily count - see Figure 1. There is also a small area at the South Pole which is not included on the map due to a lack of images. Each grid square is 120 pixels by 120 pixels, or approximately 36km at the equator - though it does vary as it approaches the poles due to the map projection. I then counted the number of craters in each square and wrote the total in the square - see Figure 2. Due to the original resolution of the map, the smallest craters visible were approximately 1.5km in diameter. Squares were only counted if they were more than 50% visible after blacking out the low-resolution areas. Craters that crossed the line between grid squares were counted towards the square that the majority of their area was in. Though included in the later density calculations, large craters greater than 1 grid square in size were not included in this initial count. They were, however, outlined in white to show their locations on the map.

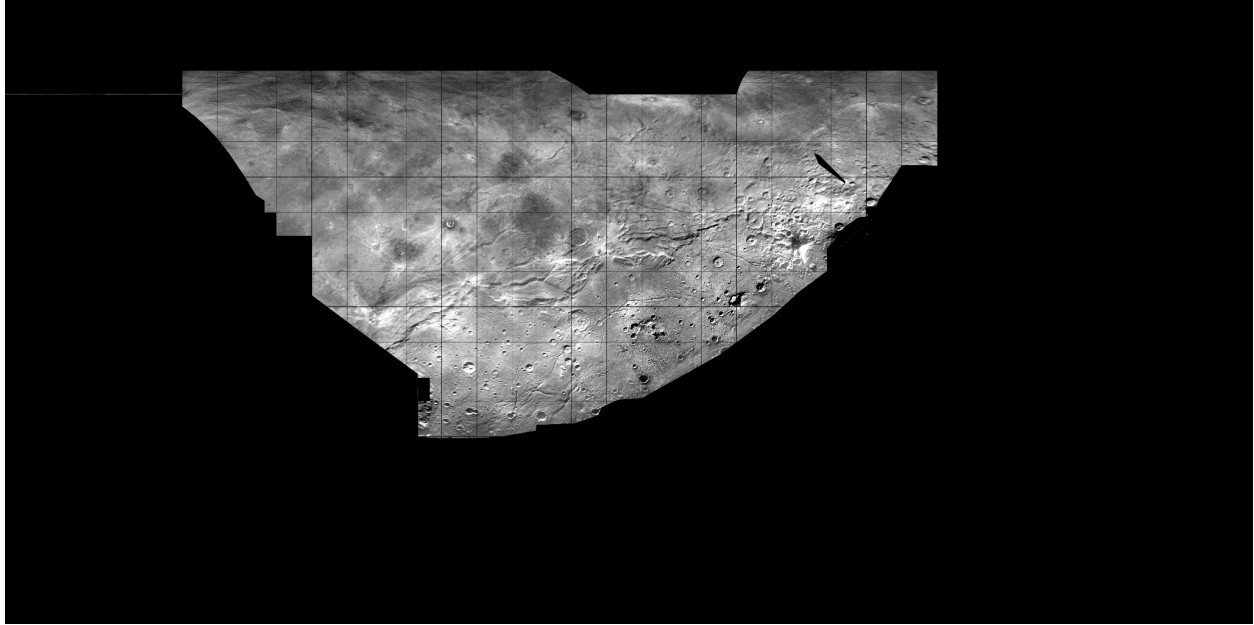


Figure 1

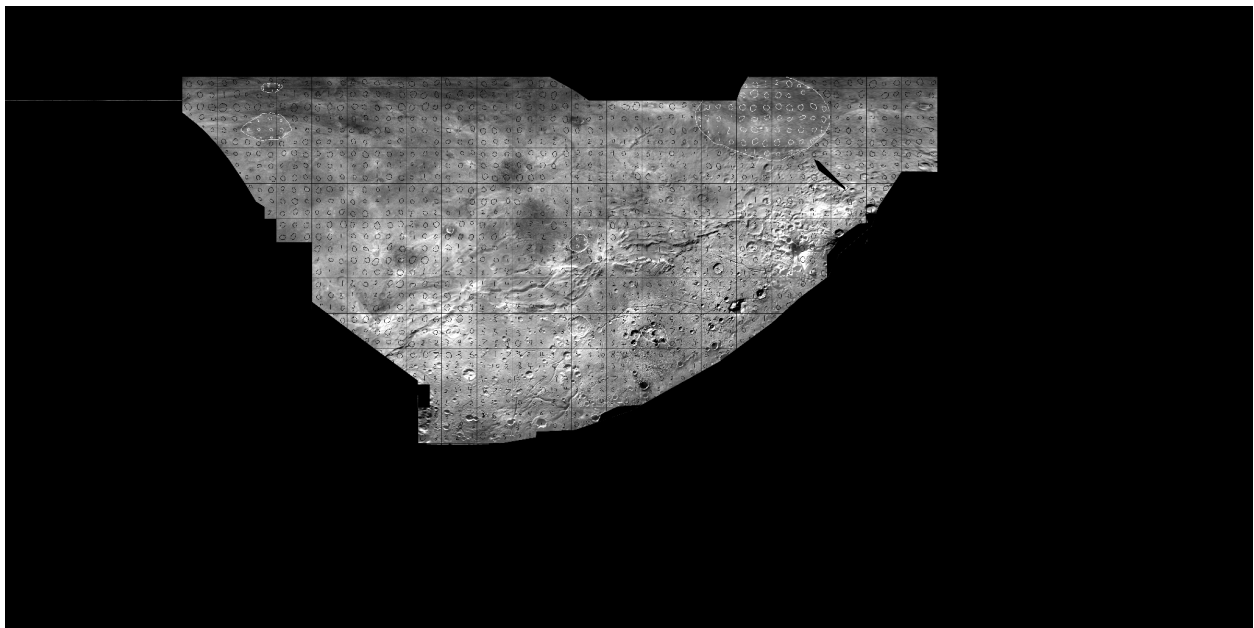


Figure 2

After counting the craters I created the crater density map by filling in and color coding each grid square based on the number of craters - darker red means more craters - see Figure 3. Then the number of grid squares with each number of craters were counted, e.g. 1 square with 15

craters and 660 with 0 craters. I used these numbers to create graphs showing the total count, northern hemisphere count, and southern hemisphere count - figures 4, 5, and 6 respectively. Those numbers were also used to calculate the average density of impact craters for the same three divisions by plugging them into the equation: $\text{\#of craters}/\text{\#of grid squares}$. The number of craters was: total 1290, northern hemisphere 1028, and southern hemisphere 262. The number of craters was: total 1708, northern hemisphere 867, and southern hemisphere 847. The average density in total was 1.324 craters/square, northern hemisphere .843 craters/square, and southern hemisphere 3.233 craters/square.

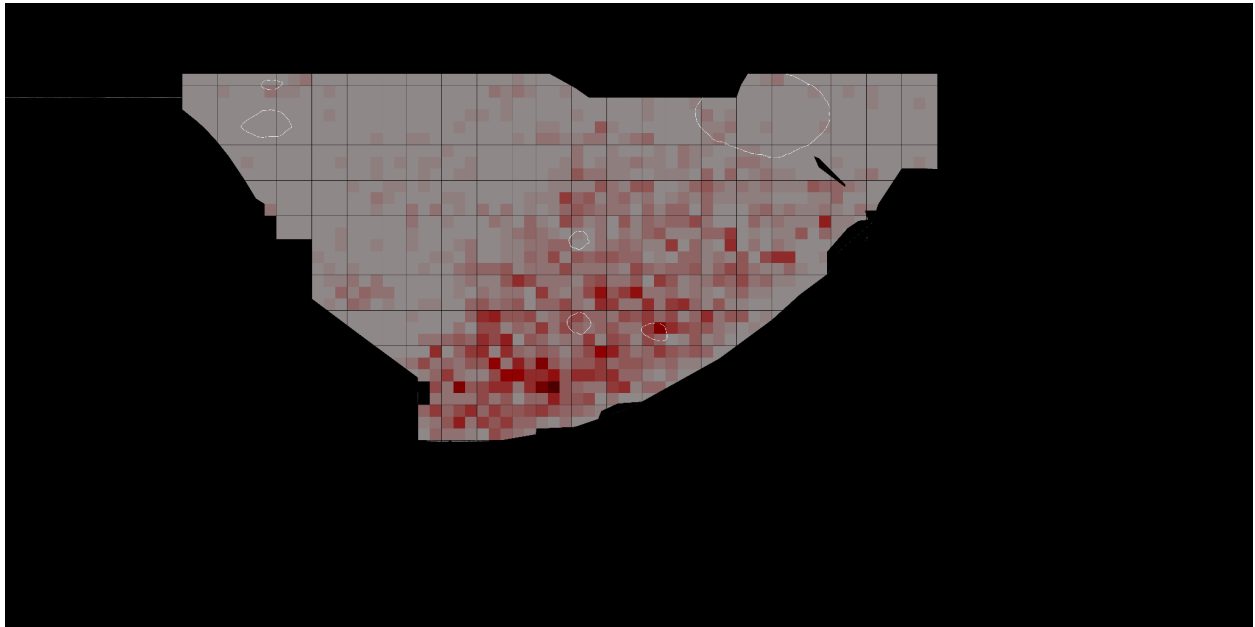


Figure 3

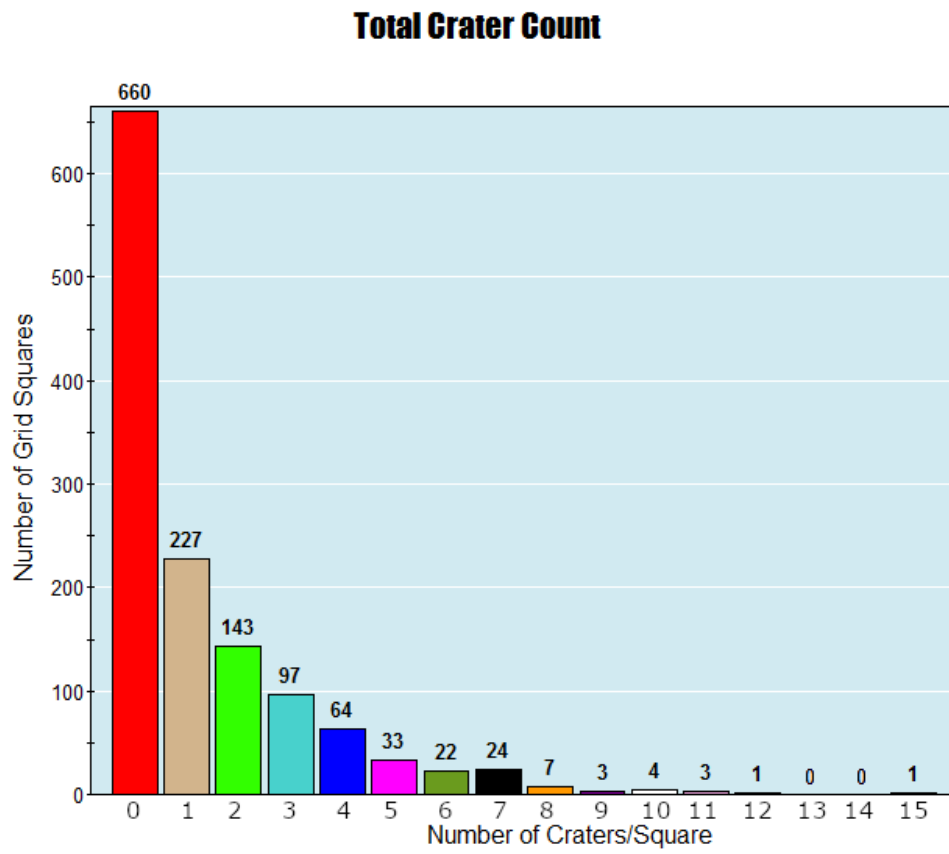


Figure 4

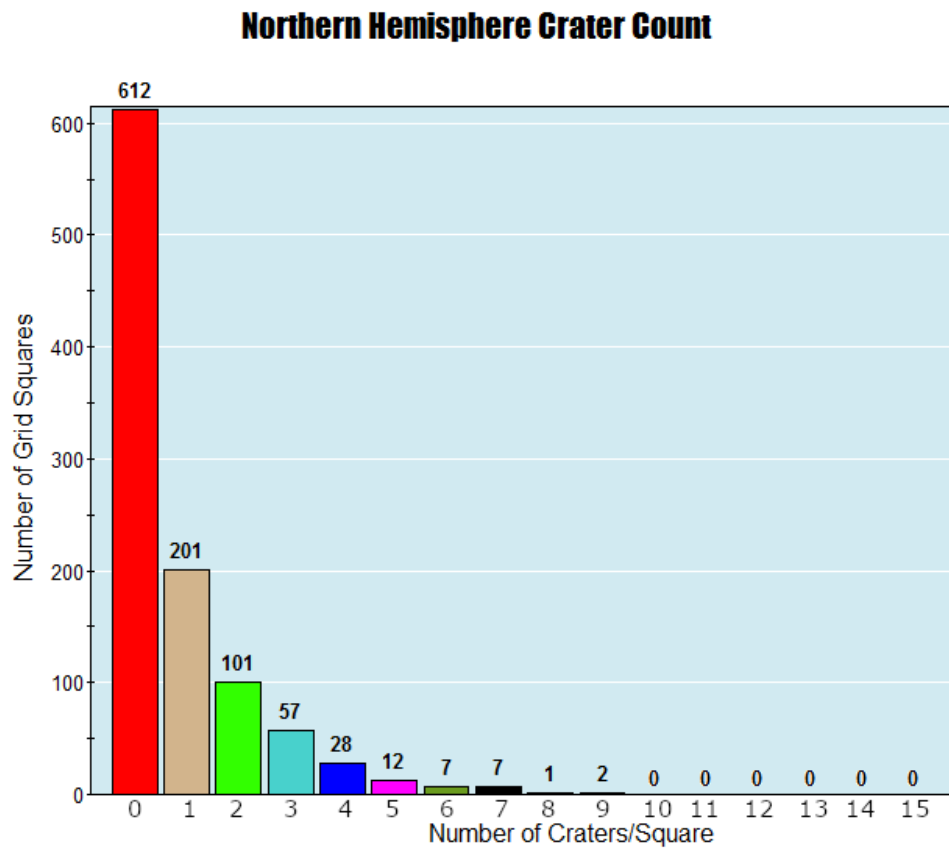


Figure 5

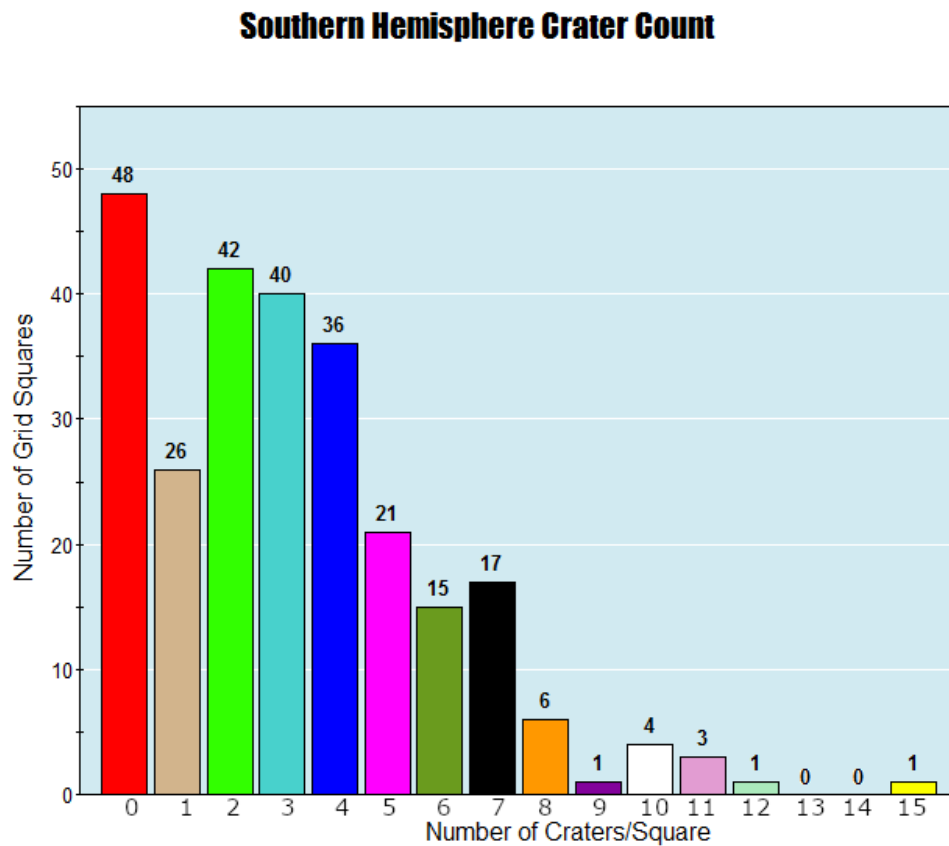


Figure 6

While the data is likely not entirely accurate due to the varying resolution of the map of Charon's surface, and the significant areas that are too low resolution to count or which we don't have images of yet it still shows some clear trends. The variation in density shows that some areas of Charon's surface are newer than others. This, combined with observations others have made of cryogeysers and cryovolcanoes (Spaceflight Now) show that Charon is likely tectonically active. It is still unknown how Charon is still active due to its small size and the nearly non-existent tidal heating from Pluto. However, one hypothesis is that the heat from radioactive decay under the surface is enough to keep things moving.

Work Cited:

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